

Sex discrimination study (1974): How we can draw causal conclusions from an experiment.

MATH 213

Sex discrimination case study

Because the experiment is blind and each manager is given only ONE file (M or F) to review, we only have information on what each manager would do in ONE of these situations. Here is what we ACTUALLY know:

Manager	Male or female file	Decision	Action if M file	Action if F file
1	M	promotion	promotion	
2	M	no promotion	no promotion	
3	M	promotion	promotion	
4	F	no promotion		no promotion
5	F	no promotion		no promotion
6	F	promotion		promotion

If we had complete information and somehow knew what decision each manager would make in EACH situation, we might have a table that looks like this:

Manager	Male or female file	Decision	Action if M file	Action if F file
1	M	promotion	promotion	promotion
2	M	no promotion	no promotion	no promotion
3	M	promotion	promotion	promotion
4	F	no promotion	no promotion	no promotion
5	F	no promotion	no promotion	no promotion
6	F	promotion	promotion	promotion

In this case, there is NO DISCRIMINATION taking place, since each manager makes the same decision, regardless of the sex associated with the file.

We can summarize this by looking at the difference of proportions:

$$\begin{aligned} &\text{Proportion of managers who promote if M file} - \text{Proportion of managers who promote if F file} \\ &= 3/6 - 3/6 = 0.5 - 0.5 = 0 \end{aligned}$$

Or, if, again, we had complete information, we might have a situation that looks like this:

Manager	Male or female file	Decision	Action if M file	Action if F file
1	M	promotion	promotion	no promotion
2	M	no promotion	no promotion	no promotion
3	M	promotion	promotion	no promotion
4	F	no promotion	promotion	no promotion
5	F	no promotion	promotion	no promotion
6	F	promotion	promotion	promotion

In this case, there IS DISCRIMINATION because SOME of the Managers opted to promote if the file was for a male but not to promote if the file was female. We summarize this again using a difference of proportions.

Proportion of managers who promote if M file - proportion of managers who promote if F file
$$= \frac{5}{6} - \frac{1}{6} = \frac{4}{6} = 0.667$$

Since the difference is POSITIVE, some of these managers are discriminating against the females.

But of course, we don't have complete information on what decision each manager would make in each situation. Here, again, is what we ACTUALLY know:

Manager	Male or female file	Decision	Action if M file	Action if F file
1	M	promotion	promotion	
2	M	no promotion	no promotion	
3	M	promotion	promotion	
4	F	no promotion		no promotion
5	F	no promotion		no promotion
6	F	promotion		promotion

So, how can we learn from this?

- Instead of looking at decisions of individual managers, we use the group of managers who were given M files to estimate the proportion of all managers who would promote if given an M file.
- Likewise, we use the group of managers who were given F files to estimate the proportion of all managers who would promote if given an F file.
- We can do this because if the groups are randomly assigned to begin with, the effect of confounding variables has been minimized and the only differences between the groups are the files that they are assigned.

Here, again, is what we ACTUALLY know:

Manager	Male or female file	Decision	Action if M file	Action if F file
1	M	promotion	promotion	
2	M	no promotion	no promotion	
3	M	promotion	promotion	
4	F	no promotion		no promotion
5	F	no promotion		no promotion
6	F	promotion		promotion

For the original example, with 24 managers in each group:

$$\begin{aligned} & (\text{Proportion of managers in male group who promote}) - (\text{proportion of managers in female group who promote}) \\ &= 21/24 - 14/24 = 7/24 = 0.29 \end{aligned}$$

Since the difference of proportions is POSITIVE, we suspect that there may be discrimination taking place. Next question: How can we be more certain that this is *caused by* the file being M or F and not due to some other reason?

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Next question: How can we be sure that this is *caused by* the file being M or F and not due to some other reason?

Note the following:

1. The subjects were RANDOMLY ASSIGNED to their groups, so the groups should tend to be balanced in terms of all characteristics.
2. But how can we be SURE that the random assignment process worked as intended to balance out the groups?

E.g. perhaps managers who are more likely to promote all ended up in the M file group by accident.

This is what we examine next!